

## HW9 adaptive signal processing I 10061622 林祐安

1.

Initial trial

$P = 50$

$\text{Alpha} = 0.05$

`% LMS Processing`

`for i = 1 : len`

`x = circshift(x, 1);`

`x(1) = x_full(i);`

`% YOUR CODE BELOW`

`y = h'*x;`

`e(i) = d(i) - y;`

`h = h + alpha*e(i)*x;`

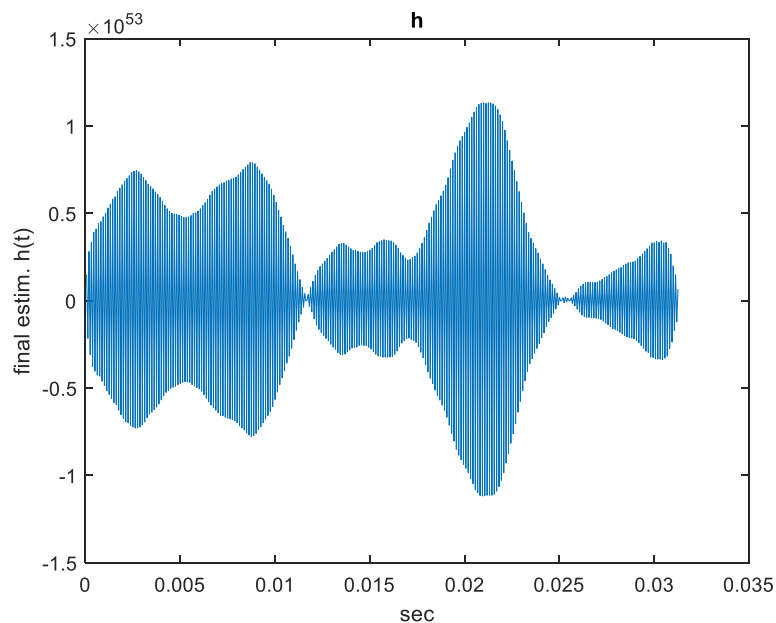
`% YOUR CODE ABOVE`

`end`

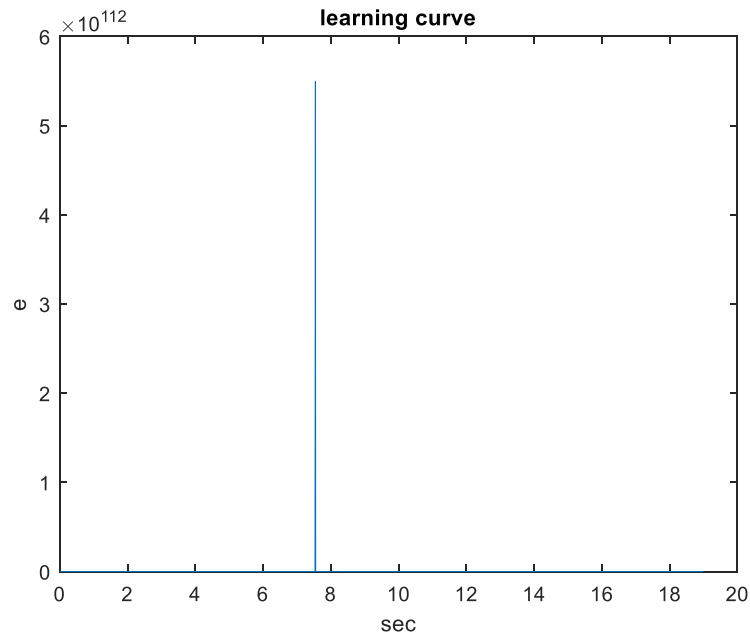
2.

**Filter length:**

Filter length affects the size of  $\mathbf{X}[n]$  and  $\mathbf{W}[n]$  discussed in class. As the filter length increases, the male's sound is suppressed much more. However, if it is too large, this algorithm would diverge like the following figure.

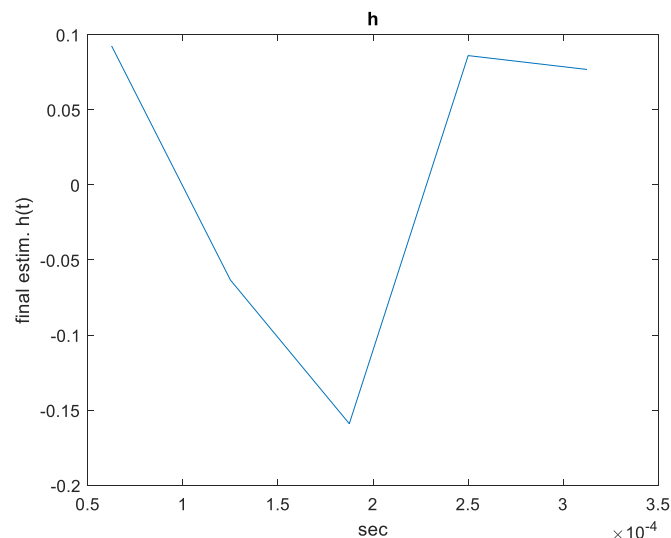


(impulse response,  $p = 500$ ,  $\alpha = 0.05$ )



(learning curve,  $p = 500$ ,  $\alpha = 0.05$ )

If we reduce the length of filter, the male's sound is louder than the original one but our goal is to obtain the female's sound as clear as better. It means this kind of adaptive filter has weak effect. Additionally, impulse response is not up to snuff seen in the following figure. In other words, longer filter length has better time resolution, hence, that contributes to more accurate impulse response.

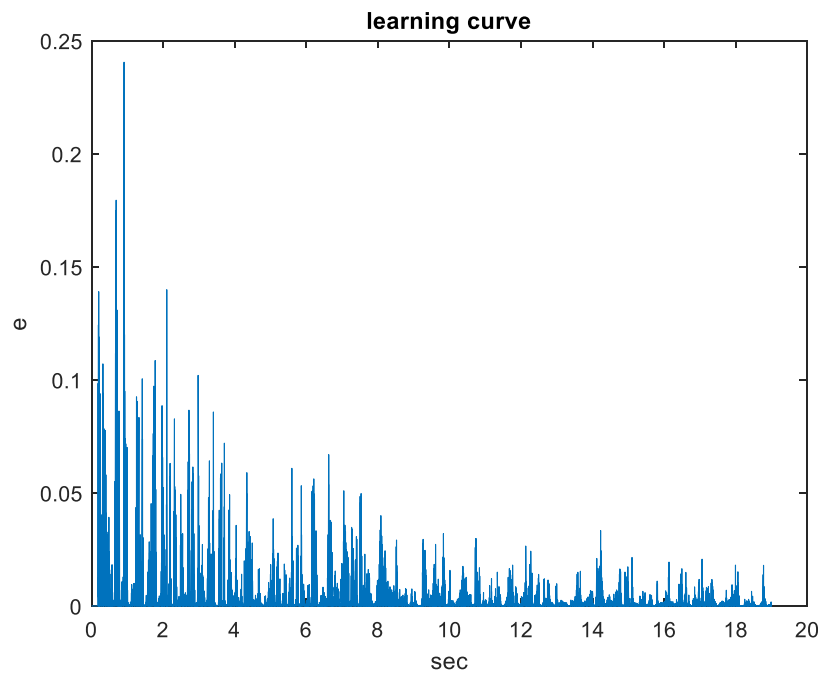


(impulse response,  $p = 5$ ,  $\alpha = 0.05$ )

It's intuitive that longer filter has better performance learned in previous homework.

### Step size:

Step size is equivalent to learning rate in machine learning. Large step size would lead algorithm to diverge, but appropriate increment of step size is worthy to speed up algorithm. In contrast, too short step size gives rise to high time consuming. Since our sound has finite length, step size also affects the performance of algorithm. Model would not be trained enough to filter desired sound due to short step size.

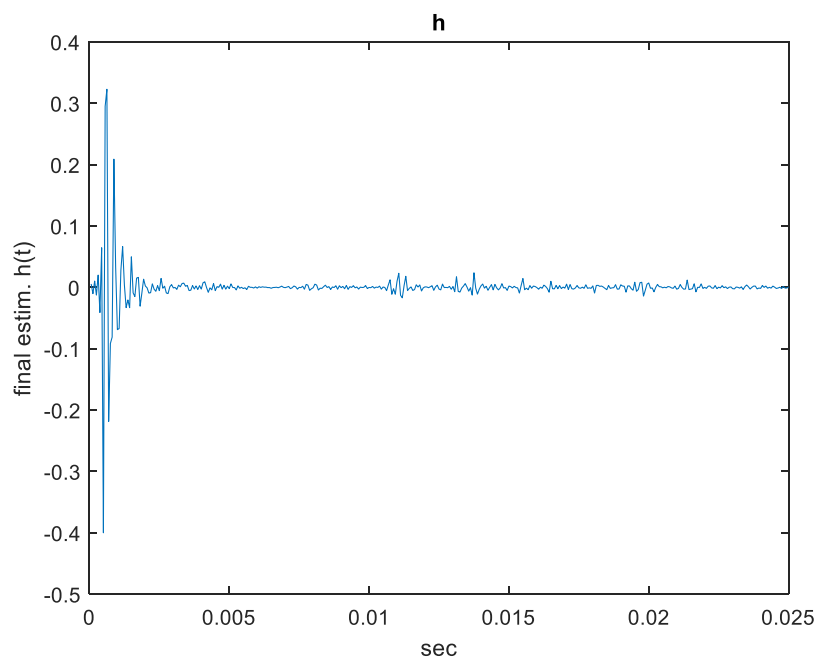


(learning curve,  $p = 50$ ,  $\alpha = 0.0005$ )

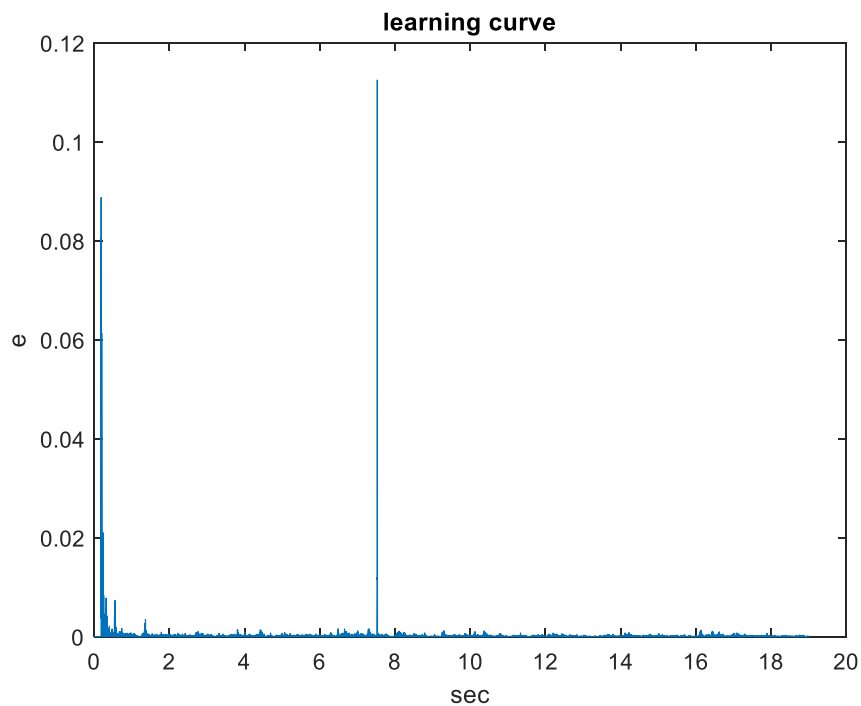
Obviously, the learning curve converges slower than the original one due to small step size. It further seems not to converge well.

After some trial and error and according to the above discussion, it has a nice work in this case,  $p = 400$  and  $\alpha = 0.05$ .

3.



(impulse response,  $p = 400$ ,  $\alpha = 0.05$ )



(learning curve,  $p = 400$ ,  $\alpha = 0.05$ )

Since step size is appropriately large, it converges fast. It goes down as time increases, that means error is efficiently reduced after updating  $h[n]$ .

4.

For  $p = 400$ , upper limit for the associated step size is about 0.051, otherwise, that results in divergence. Maybe we can determine step size automatically. For instance, step size is set about 0.1 initially and attenuates in a loop until this algorithm would not diverge.

5.

Seen in problem 2.